Probability Review

Random Variables, Expectation, and Limit Theorems

Phase 0: Prerequisites Review | Document 3 of 3
Primary texts: Ross, *A First Course in Probability*; Blitzstein & Hwang, *Introduction to Probability*

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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Abstract

This document is a self-contained refresher of classical (non-measure-theoretic) probability theory. We cover sample spaces, conditional probability, random variables, expectation, variance, joint distributions, moment generating functions, concentration inequalities, and limit theorems. Every result is stated precisely and proved completely. The treatment is elementary: we use probability mass functions, probability density functions, sums, and integrals — not σ-algebras or Lebesgue integration. Phase 01 will rebuild the entire theory on measure-theoretic foundations; the goal here is to develop fluent computational facility with the classical tools.

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1 Introduction

This document is a rigorous refresher of classical probability at the undergraduate level. It is designed for readers who have seen this material before but need to recall the key definitions, theorem statements, and proof techniques before proceeding to the measure-theoretic treatment in Phase 01.

What this document covers.

1. **Sample spaces and events** — axioms of probability, inclusion-exclusion, continuity of probability.
2. **Conditional probability and independence** — Bayes' theorem, mutual independence.
3. **Discrete random variables** — PMFs, Bernoulli, Binomial, Geometric, Poisson.
4. **Continuous random variables** — PDFs, Uniform, Exponential, Normal.
5. **Expectation** — linearity, LOTUS, indicator random variables.
6. **Variance and covariance** — computational formulas, variance of sums.
7. **Joint distributions** — marginals, conditional distributions, laws of total expectation and total variance.
8. **Moment generating functions** — computing moments, sums of independent random variables.
9. **Probability inequalities** — Markov, Chebyshev, Cauchy–Schwarz, Jensen.
10. **Limit theorems** — weak law of large numbers, central limit theorem.
11. **Tail bounds and confidence intervals** — Hoeffding's inequality, sample complexity.

What this document does not cover. We do not use σ-algebras, measurability, Lebesgue integration, filtrations, or martingale theory. These are developed from scratch in Phase 01.

Connection to Reinforcement Learning

Reinforcement learning is fundamentally about decision-making under uncertainty. Every concept here — conditional probability, expectation, variance, the law of large numbers — has a direct RL counterpart. Rewards are random variables. Value functions are conditional expectations. Monte Carlo methods converge by the law of large numbers. This review ensures you can compute fluently before Phase 01 introduces the measure-theoretic machinery that makes everything rigorous.

2 Sample Spaces and Events

Definition 2.1 (Sample Space, Outcome, Event)**.** A *sample space* Ω is a non-empty set whose elements $\omega \in \Omega$ are called *outcomes*. An *event* is a subset $A \subseteq \Omega$.

Remark 2.1. At this elementary level, we take the collection of all events to be the power set 2^Ω (all subsets of Ω). In Phase 01, this will be replaced by a σ-algebra $\mathcal{F} \subseteq 2^\Omega$, which is necessary for uncountable sample spaces.

Definition 2.2 (Set Operations on Events)**.** Let $A, B \subseteq \Omega$.

1. *Union:* $A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ ("A or B occurs").
2. *Intersection:* $A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ ("A and B occur").
3. *Complement:* $A^c = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ ("A does not occur").
4. *Disjoint:* A and B are *disjoint* (or *mutually exclusive*) if $A \cap B = \emptyset$.

2.1 Axioms of Probability

Definition 2.3 (Probability (Kolmogorov Axioms))**.** A *probability* on a sample space Ω is a function $\mathbb{P}$ assigning to each event $A \subseteq \Omega$ a real number $\mathbb{P}(A)$ satisfying:

1. **Non-negativity:** $\mathbb{P}(A) \geq 0$ for every event A.
2. **Normalization:** $\mathbb{P}(\Omega) = 1$.
3. **Countable additivity:** If $A_1, A_2, \ldots$ are pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

2.2 Basic Properties

Proposition 2.1 (Complement Rule)**.** *For any event A, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.*

Proof. Since A and A^c are disjoint and $A \cup A^c = \Omega$, Axiom 3 gives $\mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$. By Axiom 2, $\mathbb{P}(\Omega) = 1$, so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. □

Corollary 2.2. $\mathbb{P}(\emptyset) = 0$.

Proof. $\emptyset = \Omega^c$, so $\mathbb{P}(\emptyset) = 1 - \mathbb{P}(\Omega) = 1 - 1 = 0$. □

Proposition 2.3 (Monotonicity)**.** *If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*

Proof. Write $B = A \cup (B \setminus A)$, where A and $B \setminus A$ are disjoint. By Axiom 3, $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$, since $\mathbb{P}(B \setminus A) \geq 0$ by Axiom 1. □

Theorem 2.4 (Inclusion-Exclusion for Two Events)**.** *For any events A and B,*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. Write $A \cup B = A \cup (B \setminus A)$, where A and $B \setminus A$ are disjoint. By Axiom 3, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Now write $B = (A \cap B) \cup (B \setminus A)$, where $A \cap B$ and $B \setminus A$ are disjoint. By Axiom 3, $\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A)$, so $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. □

Theorem 2.5 (Inclusion-Exclusion for n Events)**.** *For events $A_1, \ldots, A_n$,*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}).$$

Proof. We proceed by induction on n.

Base case ($n = 1$): $\mathbb{P}(A_1) = \mathbb{P}(A_1)$. Trivially holds.

Base case ($n = 2$): This is Theorem 2.4.

Inductive step: Assume the formula holds for $n-1$ events. Write $\bigcup_{i=1}^{n} A_i = \left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n$. By Theorem 2.4,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right).$$

Since $\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n = \bigcup_{i=1}^{n-1} (A_i \cap A_n)$, we can apply the inductive hypothesis to both $\mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right)$ and $\mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right)$. Applying the inductive hypothesis to the first term gives the alternating sum over subsets of $\{1, \ldots, n-1\}$. Adding $\mathbb{P}(A_n)$ accounts for the singleton $\{n\}$. Subtracting the inductive hypothesis applied to $\bigcup_{i=1}^{n-1} (A_i \cap A_n)$ contributes exactly the terms involving A_n in subsets of size ≥ 2, each with the correct sign $(-1)^{k+1}$. Collecting all terms yields the formula for n events. □

2.3 Continuity of Probability

Theorem 2.6 (Continuity of Probability). *Let $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ be an increasing sequence of events. Then*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

Proof. Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The events $B_1, B_2, \ldots$ are pairwise disjoint (since $A_n \setminus A_{n-1}$ contains only elements in A_n but not in A_{n-1}, and these sets are nested), and $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$. By countable additivity,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n).$$

Now, $\sum_{k=1}^{n} \mathbb{P}(B_k) = \mathbb{P}(\bigcup_{k=1}^{n} B_k) = \mathbb{P}(A_n)$, where the first equality uses finite additivity (the B_k are disjoint) and the second uses $\bigcup_{k=1}^{n} B_k = A_n$. Therefore

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

3 Conditional Probability and Independence

3.1 Conditional Probability

Definition 3.1 (Conditional Probability). If $\mathbb{P}(B) > 0$, the *conditional probability of A given B* is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Theorem 3.1 (Multiplication Rule). *For events $A_1, \ldots, A_n$ with $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$,*

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Proof. By induction on n. For $n = 2$, the definition of conditional probability gives $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)$.

Assume the result holds for $n - 1$ events. Then

$$\begin{aligned}\mathbb{P}(A_1 \cap \cdots \cap A_n) &= \mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) \cdot \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \\ &= \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2}) \cdot \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}),\end{aligned}$$

where the second line applies the inductive hypothesis. $\square$

Theorem 3.2 (Law of Total Probability). *Let $B_1, B_2, \ldots, B_n$ be a partition of Ω (i.e., the B_i are pairwise disjoint and $\bigcup_{i=1}^{n} B_i = \Omega$) with $\mathbb{P}(B_i) > 0$ for all i. Then for any event A,*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

Proof. Since the B_i partition Ω, we have $A = A \cap \Omega = A \cap (\bigcup_{i=1}^{n} B_i) = \bigcup_{i=1}^{n} (A \cap B_i)$. The events $A \cap B_1, \ldots, A \cap B_n$ are pairwise disjoint (since the B_i are). By finite additivity,

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i),$$

where the last step uses $\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$. $\square$

Theorem 3.3 (Bayes' Theorem)**.** *Let $B_1, \ldots, B_n$ be a partition of Ω with $\mathbb{P}(B_i) > 0$ for all i, and let A be an event with $\mathbb{P}(A) > 0$. Then*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}.$$

Proof. By definition, $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j)/\mathbb{P}(A)$. The numerator equals $\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)$ by the definition of conditional probability. The denominator equals $\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$ by the law of total probability (Theorem 3.2). □

3.2 Independence

Definition 3.2 (Independence of Two Events)**.** Events A and B are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Remark 3.1. When $\mathbb{P}(B) > 0$, independence is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: knowing that B occurred does not change the probability of A.

Definition 3.3 (Mutual Independence)**.** Events $A_1, \ldots, A_n$ are *mutually independent* if for every subset $\{i_1, \ldots, i_k\} \subseteq \{1, \ldots, n\}$ with $k \geq 2$,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

This requires the product formula to hold for *all* $2^n - n - 1$ subsets of size ≥ 2, not merely for the $\binom{n}{2}$ pairs.

Pairwise $\neq$ Mutual Independence

Pairwise independence does not imply mutual independence. The following example shows that the product rule can hold for every pair yet fail for the triple.

Example 3.1 (Pairwise but Not Mutually Independent)**.** Toss two fair coins. Let $A = \{\text{first coin is heads}\}$, $B = \{\text{second coin is heads}\}$, and $C = \{\text{exactly one head}\}$. The sample space is $\{HH, HT, TH, TT\}$, each with probability $1/4$.

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$.

- $A \cap B = \{HH\}$, so $\mathbb{P}(A \cap B) = 1/4 = \mathbb{P}(A)\mathbb{P}(B)$.
- $A \cap C = \{HT\}$, so $\mathbb{P}(A \cap C) = 1/4 = \mathbb{P}(A)\mathbb{P}(C)$.
- $B \cap C = \{TH\}$, so $\mathbb{P}(B \cap C) = 1/4 = \mathbb{P}(B)\mathbb{P}(C)$.

All three pairs are independent. However, $A \cap B \cap C = \emptyset$, so $\mathbb{P}(A \cap B \cap C) = 0 \neq 1/8 = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Hence A, B, C are pairwise independent but *not* mutually independent.

Connection to RL

The Markov property — the foundation of MDPs — is an independence statement: the future is independent of the past given the present. Bayes' theorem is the update rule in Bayesian reinforcement learning and POMDP belief updates.

4 Discrete Random Variables

Definition 4.1 (Random Variable)**.** A *random variable* is a function $X \colon \Omega \to \mathbb{R}$ that assigns a real number to each outcome. We write $\{X = x\}$ as shorthand for $\{\omega \in \Omega : X(\omega) = x\}$.

Remark 4.1. At this level, the definition is purely informal: any function from Ω to $\mathbb{R}$. Phase 01 will add the measurability requirement that makes this rigorous for uncountable sample spaces.

Definition 4.2 (Probability Mass Function)**.** A random variable X is *discrete* if it takes values in a finite or countable set $\{x_1, x_2, \ldots\}$. Its *probability mass function (PMF)* is

$$p_X(x) = \mathbb{P}(X = x).$$

The PMF satisfies $p_X(x) \geq 0$ for all x and $\sum_x p_X(x) = 1$.

Definition 4.3 (Cumulative Distribution Function)**.** The *cumulative distribution function (CDF)* of a random variable X is

$$F_X(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

For a discrete random variable, $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$.

4.1 Common Discrete Distributions

Definition 4.4 (Bernoulli Distribution)**.** A random variable X has the Bernoulli(p) distribution, $0 \leq p \leq 1$, if X takes values in $\{0, 1\}$ with

$$p_X(1) = p, \quad p_X(0) = 1 - p.$$

Proposition 4.1. *If* $X \sim \text{Bernoulli}(p)$*, then* $\mathbb{E}[X] = p$ *and* $\text{Var}(X) = p(1-p)$.

Proof. $\mathbb{E}[X] = 0 \cdot (1-p) + 1 \cdot p = p$. For the variance: $\mathbb{E}[X^2] = 0^2 \cdot (1-p) + 1^2 \cdot p = p$, so $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p)$. □

Definition 4.5 (Binomial Distribution)**.** Suppose we perform n independent Bernoulli trials, each with success probability p. Let X count the total number of successes. Then $X \sim \text{Binomial}(n, p)$.

Proposition 4.2 (Binomial PMF)**.** *If* $X \sim \text{Binomial}(n, p)$*, then*

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Proof. Each specific sequence of k successes and $n-k$ failures has probability $p^k(1-p)^{n-k}$ by independence. There are $\binom{n}{k}$ such sequences (choosing which k of the n trials are successes), so $\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. □

Proposition 4.3. *If* $X \sim \text{Binomial}(n, p)$*, then* $\mathbb{E}[X] = np$ *and* $\text{Var}(X) = np(1-p)$.

Proof. Write $X = X_1 + \cdots + X_n$, where $X_i \sim \text{Bernoulli}(p)$ independently. By linearity of expectation (Theorem 6.2 below), $\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = np$. Since the X_i are independent, $\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = np(1-p)$ (using Corollary 7.7 below). □

Definition 4.6 (Geometric Distribution)**.** Perform independent Bernoulli trials with success probability $p > 0$. Let X be the trial number of the first success. Then $X \sim \text{Geometric}(p)$.

Proposition 4.4 (Geometric PMF)**.** *If* $X \sim \text{Geometric}(p)$*, then*

$$p_X(k) = (1-p)^{k-1} p, \quad k = 1, 2, 3, \ldots$$

Proof. The first success occurs on trial k exactly when the first $k-1$ trials are failures and trial k is a success. By independence, this probability is $(1-p)^{k-1} \cdot p$. □

Proposition 4.5. *If* $X \sim \text{Geometric}(p)$*, then* $\mathbb{E}[X] = 1/p$ *and* $\text{Var}(X) = (1-p)/p^2$.

Proof. Let $q = 1 - p$. We compute $\mathbb{E}[X]$ directly:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k\, q^{k-1} p = p \sum_{k=1}^{\infty} k\, q^{k-1}.$$

Recall that $\sum_{k=1}^{\infty} k\, q^{k-1} = \frac{d}{dq} \sum_{k=0}^{\infty} q^k = \frac{d}{dq} \frac{1}{1-q} = \frac{1}{(1-q)^2} = \frac{1}{p^2}$ for $|q| < 1$. Thus $\mathbb{E}[X] = p/p^2 = 1/p$.

For the variance, compute $\mathbb{E}[X^2]$. We use $\mathbb{E}[X(X-1)] = \sum_{k=1}^{\infty} k(k-1)q^{k-1}p = p\, q \sum_{k=2}^{\infty} k(k-1)q^{k-2}$. Since $\sum_{k=2}^{\infty} k(k-1)q^{k-2} = \frac{d^2}{dq^2} \frac{1}{1-q} = \frac{2}{(1-q)^3} = \frac{2}{p^3}$, we get $\mathbb{E}[X(X-1)] = 2q/p^2$. Then $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = 2q/p^2 + 1/p$, and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q + p - 1}{p^2} = \frac{q}{p^2} = \frac{1-p}{p^2},$$

where we used $2q + p - 1 = 2(1-p) + p - 1 = 1 - p = q$. □

Definition 4.7 (Poisson Distribution)**.** A random variable X has the Poisson(λ) distribution, $\lambda > 0$, if

$$p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Theorem 4.6 (Poisson as a Limit of Binomials)**.** *Let $X_n \sim \mathrm{Binomial}(n, \lambda/n)$ with $\lambda > 0$ fixed. Then for each $k = 0, 1, 2, \ldots$,*

$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}.$$

Proof. With $p = \lambda/n$,

$$\mathbb{P}(X_n = k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

Write $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!}$, so

$$\mathbb{P}(X_n = k) = \frac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

As $n \to \infty$: $\frac{n(n-1)\cdots(n-k+1)}{n^k} \to 1$ (each factor $\frac{n-j}{n} \to 1$), and $\left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}$, and $\left(1 - \frac{\lambda}{n}\right)^{-k} \to 1$. Therefore $\mathbb{P}(X_n = k) \to e^{-\lambda}\lambda^k/k!$. □

Proposition 4.7. *If $X \sim \mathrm{Poisson}(\lambda)$, then $\mathbb{E}[X] = \lambda$ and $\mathrm{Var}(X) = \lambda$.*

Proof.

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

For the variance, compute $\mathbb{E}[X(X-1)]$:

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) \frac{e^{-\lambda}\lambda^k}{k!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda^2.$$

Then $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = \lambda^2 + \lambda$, so $\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. □

5 Continuous Random Variables

Definition 5.1 (Probability Density Function)**.** A random variable X is *continuous* if there exists a non-negative function $f_X \colon \mathbb{R} \to [0, \infty)$ such that for every interval $[a, b]$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

The function f_X is called the *probability density function (PDF)* of X. The PDF satisfies $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$.

Proposition 5.1. *If X is continuous with PDF f_X, then its CDF is $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$, and $f_X(x) = F_X'(x)$ wherever F_X is differentiable.*

Proof. By definition, $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt$. By the fundamental theorem of calculus, $F_X'(x) = f_X(x)$ at every point where f_X is continuous. □

Proposition 5.2. *If X is a continuous random variable, then $\mathbb{P}(X = x) = 0$ for every $x \in \mathbb{R}$.*

Proof. $\mathbb{P}(X = x) = \mathbb{P}(x \leq X \leq x) = \int_x^x f_X(t)\, dt = 0$. □

5.1 Common Continuous Distributions

Definition 5.2 (Uniform Distribution)**.** $X \sim \text{Uniform}(a, b)$ if X has PDF

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 5.3. *If $X \sim \text{Uniform}(a, b)$, then $\mathbb{E}[X] = \frac{a+b}{2}$ and $\text{Var}(X) = \frac{(b-a)^2}{12}$.*

Proof.

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \cdot \frac{x^2}{2}\bigg|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \cdot \frac{x^3}{3}\bigg|_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}.$$

Thus

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} = \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12}.$$

□

Definition 5.3 (Exponential Distribution)**.** $X \sim \text{Exponential}(\lambda)$, $\lambda > 0$, if X has PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Proposition 5.4. *If $X \sim \text{Exponential}(\lambda)$, then $\mathbb{E}[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.*

Proof. Using integration by parts with $u = x$, $dv = \lambda e^{-\lambda x}\, dx$:

$$\mathbb{E}[X] = \int_0^\infty x \lambda e^{-\lambda x}\, dx = \left[-x e^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\, dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

Similarly, integrating by parts twice:

$$\mathbb{E}[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x}\, dx = \frac{2}{\lambda^2}.$$

Thus $\text{Var}(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. □

Theorem 5.5 (Memoryless Property of the Exponential). *If $X \sim \text{Exponential}(\lambda)$, then for all $s, t \geq 0$,*

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Proof. The survival function is $\mathbb{P}(X > x) = e^{-\lambda x}$ for $x \geq 0$. By definition of conditional probability,

$$\mathbb{P}(X > s{+}t \mid X > s) = \frac{\mathbb{P}(X > s + t \text{ and } X > s)}{\mathbb{P}(X > s)} = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t).$$

□

Definition 5.4 (Normal (Gaussian) Distribution). $X \sim \mathcal{N}(\mu, \sigma^2)$ if X has PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

When $\mu = 0$ and $\sigma^2 = 1$, we write $Z \sim \mathcal{N}(0, 1)$ and call Z a *standard normal* random variable.

Theorem 5.6 (The Gaussian Integral). $\displaystyle\int_{-\infty}^{\infty} e^{-x^2/2}\, dx = \sqrt{2\pi}$. *Consequently, the standard normal PDF integrates to* 1.

Proof. Let $I = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$. We show $I^2 = 2\pi$. Consider

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\, dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\, dx\, dy.$$

Switch to polar coordinates: $x = r\cos\theta$, $y = r\sin\theta$, $dx\, dy = r\, dr\, d\theta$:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\, r\, dr\, d\theta = 2\pi \int_0^{\infty} r\, e^{-r^2/2}\, dr.$$

Substituting $u = r^2/2$, $du = r\, dr$:

$$\int_0^{\infty} r\, e^{-r^2/2}\, dr = \int_0^{\infty} e^{-u}\, du = 1.$$

Therefore $I^2 = 2\pi$, so $I = \sqrt{2\pi}$ (since $I > 0$). It follows that $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = 1$. □

Proposition 5.7. *If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.*

Proof. Let $Z = (X - \mu)/\sigma$, so $Z \sim \mathcal{N}(0, 1)$ and $X = \mu + \sigma Z$. It suffices to show $\mathbb{E}[Z] = 0$ and $\text{Var}(Z) = 1$.

For the mean: $\mathbb{E}[Z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z\, e^{-z^2/2}\, dz = 0$, since the integrand $z \mapsto z\, e^{-z^2/2}$ is an odd function.

For the variance: $\mathbb{E}[Z^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz$. Integrate by parts with $u = z$, $dv = z\, e^{-z^2/2}\, dz$, so $du = dz$ and $v = -e^{-z^2/2}$:

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz = \left[-z\, e^{-z^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\, dz = 0 + \sqrt{2\pi}.$$

Thus $\mathbb{E}[Z^2] = 1$, and $\text{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = 1 - 0 = 1$. Therefore $\mathbb{E}[X] = \mu + \sigma \cdot 0 = \mu$ and $\text{Var}(X) = \sigma^2\, \text{Var}(Z) = \sigma^2$. □

6 Expectation

Definition 6.1 (Expectation)**.** Let X be a random variable.

- **Discrete case:** If X is discrete with PMF p_X, then
$$\mathbb{E}[X] = \sum_x x\, p_X(x),$$
provided the sum converges absolutely.
- **Continuous case:** If X is continuous with PDF f_X, then
$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx,$$
provided the integral converges absolutely.

Theorem 6.1 (Law of the Unconscious Statistician (LOTUS))**.** *Let $g\colon \mathbb{R} \to \mathbb{R}$.*

- ***Discrete:*** $\mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x)$.
- ***Continuous:*** $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$.

Proof (discrete case). Let $Y = g(X)$. For each value y in the range of g, $p_Y(y) = \mathbb{P}(Y = y) = \sum_{x:g(x)=y} p_X(x)$. Then

$$\mathbb{E}[Y] = \sum_y y\, p_Y(y) = \sum_y y \sum_{x:g(x)=y} p_X(x) = \sum_y \sum_{x:g(x)=y} g(x)\, p_X(x) = \sum_x g(x)\, p_X(x),$$

where the last step rearranges the double sum: every x appears exactly once in the partition $\{x : g(x) = y\}$ as y ranges over the values of g. □

Theorem 6.2 (Linearity of Expectation)**.** *For any random variables X and Y (on the same sample space) and constants $a, b \in \mathbb{R}$,*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

*This holds **without any independence assumption**.*

Proof. We prove the discrete case; the continuous case is analogous with integrals.

First, $\mathbb{E}[aX] = \sum_x (ax)\, p_X(x) = a \sum_x x\, p_X(x) = a\mathbb{E}[X]$.

For the sum: let (X, Y) have joint PMF $p_{X,Y}$. Then

$$\begin{aligned}
\mathbb{E}[X + Y] &= \sum_x \sum_y (x + y)\, p_{X,Y}(x, y) \\
&= \sum_x \sum_y x\, p_{X,Y}(x, y) + \sum_x \sum_y y\, p_{X,Y}(x, y) \\
&= \sum_x x \sum_y p_{X,Y}(x, y) + \sum_y y \sum_x p_{X,Y}(x, y) \\
&= \sum_x x\, p_X(x) + \sum_y y\, p_Y(y) = \mathbb{E}[X] + \mathbb{E}[Y].
\end{aligned}$$

Combining: $\mathbb{E}[aX + bY] = \mathbb{E}[aX] + \mathbb{E}[bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. □

Proposition 6.3 (Expectation of Indicator Random Variables)**.** *Let A be an event, and define $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ otherwise. Then $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.*

Proof. $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$, so $\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot (1 - \mathbb{P}(A)) = \mathbb{P}(A)$. □

Linearity of Expectation

Linearity of expectation is arguably the single most useful property in probability. It holds unconditionally — no independence required. In RL, the value function $V^\pi(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s\right]$ is linear in the reward, and linearity lets us decompose it step by step.

7 Variance and Covariance

Definition 7.1 (Variance)**.** The *variance* of a random variable X is

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right],$$

provided this expectation exists.

Theorem 7.1 (Computational Formula for Variance)**.** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Proof. Let $\mu = \mathbb{E}[X]$. Expanding the square:

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

□

Proposition 7.2 (Affine Transformation of Variance)**.** *For constants $a, b \in \mathbb{R}$,* $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

Proof. $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$, so

$$\mathrm{Var}(aX+b) = \mathbb{E}[(aX+b-a\mathbb{E}[X]-b)^2] = \mathbb{E}[a^2(X-\mathbb{E}[X])^2] = a^2\mathbb{E}[(X-\mathbb{E}[X])^2] = a^2\,\mathrm{Var}(X).$$ □

Definition 7.2 (Standard Deviation)**.** The *standard deviation* of X is $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

Definition 7.3 (Covariance)**.** The *covariance* of random variables X and Y is

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Theorem 7.3 (Computational Formula for Covariance)**.** $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Expanding:

$$\begin{aligned}\mathrm{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY - \mu_Y X - \mu_X Y + \mu_X\mu_Y] \\ &= \mathbb{E}[XY] - \mu_Y\mathbb{E}[X] - \mu_X\mathbb{E}[Y] + \mu_X\mu_Y = \mathbb{E}[XY] - \mu_X\mu_Y.\end{aligned}$$ □

Proposition 7.4 (Properties of Covariance)**.** *For random variables X, Y, Z and constants $a, b \in \mathbb{R}$:*

1. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
2. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$ *(symmetry).*
3. $\mathrm{Cov}(aX + bY, Z) = a\,\mathrm{Cov}(X, Z) + b\,\mathrm{Cov}(Y, Z)$ *(bilinearity).*

Proof. **(1)** $\mathrm{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}(X)$.
(2) Immediate from $\mathbb{E}[XY] = \mathbb{E}[YX]$.
(3) Using the computational formula:

$$\begin{aligned}\mathrm{Cov}(aX + bY, Z) &= \mathbb{E}[(aX + bY)Z] - \mathbb{E}[aX + bY]\mathbb{E}[Z] \\ &= a\mathbb{E}[XZ] + b\mathbb{E}[YZ] - (a\mathbb{E}[X] + b\mathbb{E}[Y])\mathbb{E}[Z] \\ &= a(\mathbb{E}[XZ] - \mathbb{E}[X]\mathbb{E}[Z]) + b(\mathbb{E}[YZ] - \mathbb{E}[Y]\mathbb{E}[Z]) \\ &= a\,\mathrm{Cov}(X, Z) + b\,\mathrm{Cov}(Y, Z).\end{aligned}$$ □

Theorem 7.5 (Variance of a Sum)**.** $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.

Proof.

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathrm{Cov}(X+Y, X+Y)\\ &= \mathrm{Cov}(X,X) + \mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,X) + \mathrm{Cov}(Y,Y)\\ &= \mathrm{Var}(X) + 2\,\mathrm{Cov}(X,Y) + \mathrm{Var}(Y),\end{aligned}$$

where the second line uses bilinearity and the third uses $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$, $\mathrm{Cov}(Y,Y) = \mathrm{Var}(Y)$, and symmetry. □

Theorem 7.6 (Independent Implies Uncorrelated)**.** *If X and Y are independent random variables, then* $\mathrm{Cov}(X,Y) = 0$.

Proof. For independent discrete random variables,

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_{X,Y}(x,y) = \sum_x \sum_y xy\, p_X(x)\, p_Y(y) = \left(\sum_x x\, p_X(x)\right)\left(\sum_y y\, p_Y(y)\right) = \mathbb{E}[X]\mathbb{E}[Y].$$

By Theorem 7.3, $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$. □

Corollary 7.7 (Variance of Sum of Independent Random Variables)**.** *If $X_1, \ldots, X_n$ are independent, then* $\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)$.

Proof. By induction using Theorem 7.5 and Theorem 7.6: all cross-covariance terms vanish. □

Definition 7.4 (Correlation Coefficient)**.** The *correlation* of X and Y (with $\mathrm{Var}(X), \mathrm{Var}(Y) > 0$) is

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

One can show $-1 \le \rho(X,Y) \le 1$ (this follows from the Cauchy–Schwarz inequality, Theorem 10.3).

7.1 Variances of the Standard Distributions

We collect the variance computations for the distributions of Sections 4 and 5. The proofs of $\mathrm{Var}(X) = p(1-p)$ for Bernoulli, $\mathrm{Var}(X) = np(1-p)$ for Binomial, $\mathrm{Var}(X) = (1-p)/p^2$ for Geometric, and $\mathrm{Var}(X) = \lambda$ for Poisson were given in those sections. Similarly, $\mathrm{Var}(X) = (b-a)^2/12$ for Uniform, $\mathrm{Var}(X) = 1/\lambda^2$ for Exponential, and $\mathrm{Var}(X) = \sigma^2$ for Normal were proved there.

Distribution	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	p	$p(1-p)$
Binomial(n,p)	np	$np(1-p)$
Geometric(p)	$1/p$	$(1-p)/p^2$
Poisson(λ)	λ	λ
Uniform(a,b)	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$1/\lambda$	$1/\lambda^2$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2

8 Joint Distributions

Definition 8.1 (Joint PMF)**.** Discrete random variables X and Y have *joint PMF*

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

Definition 8.2 (Joint PDF)**.** Continuous random variables X and Y have *joint PDF* $f_{X,Y}$ if

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\, dx\, dy$$

for every "reasonable" region $A \subseteq \mathbb{R}^2$.

8.1 Marginal Distributions

Proposition 8.1 (Marginal from Joint)**.**
- ***Discrete:*** $p_X(x) = \sum_y p_{X,Y}(x, y)$.
- ***Continuous:*** $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$.

Proof. In the discrete case, $p_X(x) = \mathbb{P}(X = x) = \mathbb{P}\Big(\bigcup_y \{X = x, Y = y\}\Big) = \sum_y \mathbb{P}(X = x, Y = y) = \sum_y p_{X,Y}(x, y)$, where the third equality uses countable additivity (the events $\{Y = y\}$ are disjoint). The continuous case is analogous with integration replacing summation. □

8.2 Conditional Distributions

Definition 8.3 (Conditional PMF)**.** If $\mathbb{P}(X = x) > 0$, the *conditional PMF of Y given $X = x$* is

$$p_{Y|X}(y \mid x) = \mathbb{P}(Y = y \mid X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

Definition 8.4 (Conditional PDF)**.** If $f_X(x) > 0$, the *conditional PDF of Y given $X = x$* is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

8.3 Independence of Random Variables

Definition 8.5 (Independence of Random Variables)**.** Random variables X and Y are *independent* if their joint distribution factors:
- **Discrete:** $p_{X,Y}(x, y) = p_X(x)\, p_Y(y)$ for all x, y.
- **Continuous:** $f_{X,Y}(x, y) = f_X(x)\, f_Y(y)$ for all x, y.

8.4 Conditional Expectation

Definition 8.6 (Conditional Expectation Given an Event)**.** In the discrete case, the *conditional expectation of Y given $X = x$* is

$$\mathbb{E}[Y \mid X = x] = \sum_y y\, p_{Y|X}(y \mid x).$$

In the continuous case,

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y\, f_{Y|X}(y \mid x)\, dy.$$

The function $g(x) = \mathbb{E}[Y \mid X = x]$ defines a new random variable $\mathbb{E}[Y \mid X] = g(X)$.

Theorem 8.2 (Law of Total Expectation (Tower Property))**.** $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$.

Proof (discrete case). Let $g(x) = \mathbb{E}[Y \mid X = x]$. Then

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[g(X)] &= \sum_x g(x)\, p_X(x) = \sum_x \left(\sum_y y\, p_{Y|X}(y \mid x) \right) p_X(x) \\ &= \sum_x \sum_y y \cdot \frac{p_{X,Y}(x,y)}{p_X(x)} \cdot p_X(x) = \sum_x \sum_y y\, p_{X,Y}(x,y) \\ &= \sum_y y \sum_x p_{X,Y}(x,y) = \sum_y y\, p_Y(y) = \mathbb{E}[Y].\end{aligned}$$

□

Theorem 8.3 (Law of Total Variance)**.** $\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$.

Proof. We use $\mathrm{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$.

The conditional variance is $\mathrm{Var}(Y \mid X) = \mathbb{E}[Y^2 \mid X] - (\mathbb{E}[Y \mid X])^2$. Taking expectations of both sides and using the tower property:

$$\mathbb{E}[\mathrm{Var}(Y \mid X)] = \mathbb{E}[\mathbb{E}[Y^2 \mid X]] - \mathbb{E}[(\mathbb{E}[Y \mid X])^2] = \mathbb{E}[Y^2] - \mathbb{E}[(\mathbb{E}[Y \mid X])^2].$$

For the second term, $\mathrm{Var}(\mathbb{E}[Y \mid X]) = \mathbb{E}[(\mathbb{E}[Y \mid X])^2] - (\mathbb{E}[\mathbb{E}[Y \mid X]])^2 = \mathbb{E}[(\mathbb{E}[Y \mid X])^2] - (\mathbb{E}[Y])^2$. Adding:

$$\begin{aligned}\mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X]) &= \left(\mathbb{E}[Y^2] - \mathbb{E}[(\mathbb{E}[Y \mid X])^2]\right) + \left(\mathbb{E}[(\mathbb{E}[Y \mid X])^2] - (\mathbb{E}[Y])^2\right) \\ &= \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \mathrm{Var}(Y).\end{aligned}$$

□

Tower Property and Dynamic Programming

The law of total expectation (tower property) is the engine of dynamic programming. The Bellman equation

$$V(s) = \mathbb{E}[R + \gamma V(S') \mid S = s] = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \left[R(s,a,s') + \gamma V(s')\right]$$

is obtained by conditioning on the action and next state. Phase 01 will prove this rigorously with σ-algebras.

9 Moment Generating Functions

Definition 9.1 (Moment Generating Function)**.** The *moment generating function (MGF)* of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

defined for all t in an open interval containing 0 where this expectation is finite.

Theorem 9.1 (MGF Determines the Distribution)**.** *If $M_X(t) = M_Y(t)$ for all t in an open interval containing 0, then X and Y have the same distribution.*

Remark 9.1. The proof uses the theory of Laplace transforms and analytic continuation. We state this theorem without proof and use it as a tool for identifying distributions.

Theorem 9.2 (Moments from MGF)**.** *If $M_X(t)$ exists in a neighborhood of 0, then*

$$\mathbb{E}[X^n] = M_X^{(n)}(0) = \frac{d^n}{dt^n} M_X(t)\bigg|_{t=0}$$

for every $n \geq 1$.

Proof. Since $e^{tX} = \sum_{k=0}^{\infty} \frac{(tX)^k}{k!}$ and the MGF exists in a neighborhood of 0 (justifying interchange of expectation and summation), we have

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} \frac{t^k}{k!}\mathbb{E}[X^k].$$

Differentiating n times with respect to t:

$$M_X^{(n)}(t) = \sum_{k=n}^{\infty} \frac{t^{k-n}}{(k-n)!}\mathbb{E}[X^k].$$

Setting $t = 0$: $M_X^{(n)}(0) = \mathbb{E}[X^n]$ (only the $k = n$ term survives). □

Theorem 9.3 (MGF of Sum of Independent Random Variables)**.** *If X and Y are independent, then $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.*

Proof. $M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} \cdot e^{tY}] = \mathbb{E}[e^{tX}] \cdot \mathbb{E}[e^{tY}] = M_X(t) \cdot M_Y(t)$, where the third equality uses the fact that for independent random variables, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ (with $g(x) = e^{tx}$ and $h(y) = e^{ty}$). □

9.1 MGFs of Standard Distributions

Proposition 9.4 (MGF of Bernoulli)**.** *If $X \sim \text{Bernoulli}(p)$, then $M_X(t) = 1 - p + pe^t$.*

Proof. $M_X(t) = \mathbb{E}[e^{tX}] = e^{t\cdot 0}(1-p) + e^{t\cdot 1}p = (1-p) + pe^t$. □

Proposition 9.5 (MGF of Poisson)**.** *If $X \sim \text{Poisson}(\lambda)$, then $M_X(t) = \exp(\lambda(e^t - 1))$.*

Proof.

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$ □

Proposition 9.6 (MGF of Normal)**.** *If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $M_X(t) = \exp(\mu t + \sigma^2 t^2/2)$.*

Proof. It suffices to prove the case $Z \sim \mathcal{N}(0,1)$, since $M_X(t) = \mathbb{E}[e^{t(\mu+\sigma Z)}] = e^{\mu t}M_Z(\sigma t)$.

For $Z \sim \mathcal{N}(0,1)$:

$$\begin{aligned} M_Z(t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{tz}e^{-z^2/2}\,dz = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(z^2-2tz)/2}\,dz \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-((z-t)^2-t^2)/2}\,dz = e^{t^2/2} \cdot \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(z-t)^2/2}\,dz = e^{t^2/2}, \end{aligned}$$

where the last integral equals 1 by the substitution $u = z - t$ and Theorem 5.6. Thus $M_X(t) = e^{\mu t} \cdot e^{(\sigma t)^2/2} = e^{\mu t + \sigma^2 t^2/2}$. □

9.2 Applications of MGFs

Theorem 9.7 (Sum of Independent Poissons)**.** *If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.*

Proof. $M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\lambda(e^t-1)} \cdot e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)}$, which is the MGF of $\text{Poisson}(\lambda + \mu)$. By Theorem 9.1, $X + Y \sim \text{Poisson}(\lambda + \mu)$. □

Theorem 9.8 (Sum of Independent Normals)**.** *If $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.*

Proof. $M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\mu_1 t + \sigma_1^2 t^2/2} \cdot e^{\mu_2 t + \sigma_2^2 t^2/2} = e^{(\mu_1+\mu_2)t + (\sigma_1^2+\sigma_2^2)t^2/2}$, which is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. The result follows from Theorem 9.1. □

10 Probability Inequalities

Theorem 10.1 (Markov's Inequality). *If $X \geq 0$ and $a > 0$, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. We have $X \geq a \cdot \mathbf{1}_{\{X \geq a\}}$, since: if $X \geq a$ then the right side is $a \leq X$, and if $X < a$ then the right side is $0 \leq X$. Taking expectations:

$$\mathbb{E}[X] \geq a \cdot \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = a \cdot \mathbb{P}(X \geq a).$$

Dividing by $a > 0$ gives the result. □

Theorem 10.2 (Chebyshev's Inequality). *For any random variable X with finite variance and any $k > 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}.$$

Proof. Apply Markov's inequality to the non-negative random variable $(X - \mathbb{E}[X])^2$ with threshold k^2:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k) = \mathbb{P}((X - \mathbb{E}[X])^2 \geq k^2) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{k^2} = \frac{\mathrm{Var}(X)}{k^2}.$$ □

Theorem 10.3 (Cauchy–Schwarz Inequality for Expectations). *For any random variables X and Y with $\mathbb{E}[X^2], \mathbb{E}[Y^2] < \infty$,*

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

Proof. If $\mathbb{E}[Y^2] = 0$, then $Y = 0$ a.s., so $\mathbb{E}[XY] = 0$ and both sides are 0.

Assume $\mathbb{E}[Y^2] > 0$. For any $t \in \mathbb{R}$, $\mathbb{E}[(X + tY)^2] \geq 0$. Expanding:

$$\mathbb{E}[X^2] + 2t\,\mathbb{E}[XY] + t^2\,\mathbb{E}[Y^2] \geq 0 \quad \text{for all } t \in \mathbb{R}.$$

This is a quadratic in t that is non-negative for all t. A quadratic $at^2 + bt + c$ with $a > 0$ is non-negative for all t if and only if its discriminant satisfies $b^2 - 4ac \leq 0$. Here $a = \mathbb{E}[Y^2] > 0$, $b = 2\mathbb{E}[XY]$, $c = \mathbb{E}[X^2]$, so

$$4(\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0,$$

which gives $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$. □

Theorem 10.4 (Jensen's Inequality). *If $g\colon \mathbb{R} \to \mathbb{R}$ is convex and $\mathbb{E}[|X|] < \infty$, then*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

Proof (discrete case, by induction). We prove that if X takes values $x_1, \ldots, x_n$ with probabilities $p_1, \ldots, p_n$ (where $p_i > 0$ and $\sum p_i = 1$), then $g(\sum_{i=1}^n p_i x_i) \leq \sum_{i=1}^n p_i\, g(x_i)$.

Base case ($n = 1$): $g(p_1 x_1) = g(x_1) = p_1 g(x_1)$. Holds trivially.

Base case ($n = 2$): This is the definition of convexity: $g(p_1 x_1 + p_2 x_2) \leq p_1 g(x_1) + p_2 g(x_2)$ with $p_1 + p_2 = 1$.

Inductive step: Assume the result for $n - 1$ values. Let $q = 1 - p_n > 0$ (the case $p_n = 1$ is trivial). Write

$$\sum_{i=1}^{n} p_i x_i = q \cdot \underbrace{\sum_{i=1}^{n-1} \frac{p_i}{q} x_i}_{=\bar{x}} + p_n x_n.$$

Since $q + p_n = 1$, convexity gives

$$g\left(\sum_{i=1}^{n} p_i x_i\right) = g(q\bar{x} + p_n x_n) \leq q\, g(\bar{x}) + p_n\, g(x_n).$$

The weights p_i/q for $i = 1, \ldots, n-1$ sum to 1, so the inductive hypothesis gives $g(\bar{x}) \leq \sum_{i=1}^{n-1} \frac{p_i}{q} g(x_i)$. Substituting:

$$g\left(\sum_{i=1}^{n} p_i x_i\right) \leq q \sum_{i=1}^{n-1} \frac{p_i}{q} g(x_i) + p_n\, g(x_n) = \sum_{i=1}^{n} p_i\, g(x_i). \qquad \square$$

Concentration Inequalities in RL

Concentration inequalities are the quantitative tools that turn convergence concepts into convergence rates. Markov and Chebyshev give basic tail bounds. Phase 01 will add the sharper Hoeffding and Azuma–Hoeffding inequalities used in RL convergence proofs.

11 Limit Theorems

Definition 11.1 (Convergence in Probability)**.** A sequence of random variables $X_1, X_2, \ldots$ *converges in probability* to a random variable X, written $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Theorem 11.1 (Weak Law of Large Numbers (WLLN))**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed (i.i.d.) random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.*

Proof. By linearity of expectation, $\mathbb{E}[\bar{X}_n] = \mu$. By independence,

$$\mathrm{Var}(\bar{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

By Chebyshev's inequality (Theorem 10.2), for any $\varepsilon > 0$:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0 \quad \text{as } n \to \infty. \qquad \square$$

Theorem 11.2 (Strong Law of Large Numbers (SLLN))**.** *Under the same hypotheses as Theorem 11.1, the convergence holds almost surely: $\bar{X}_n \to \mu$ with probability* 1.

Remark 11.1. The SLLN requires measure-theoretic tools (the Borel–Cantelli lemma and truncation arguments) and is proved in Phase 01.

Definition 11.2 (Convergence in Distribution)**.** A sequence $X_1, X_2, \ldots$ *converges in distribution* to X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

at every point x where F_X is continuous.

Theorem 11.3 (Central Limit Theorem (CLT))**.** *Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 \in (0, \infty)$. Then*

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Remark 11.2. The standard proof uses characteristic functions (or equivalently, MGFs when they exist) and is deferred to Phase 01. Here we state the result and illustrate its use.

Example 11.1 (CLT Application: Polling)**.** A poll surveys $n = 1000$ voters, each independently supporting a candidate with unknown probability p. The sample proportion is $\hat{p} = \bar{X}_n$ where $X_i \sim \text{Bernoulli}(p)$. By the CLT,

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx \mathcal{N}(0, 1).$$

A 95% confidence interval for p is approximately $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$. With $n = 1000$ and $\hat{p} = 0.52$, the margin of error is $1.96\sqrt{0.52 \cdot 0.48/1000} \approx 0.031$.

Example 11.2 (CLT Application: Sum of Dice)**.** Roll a fair die $n = 100$ times and let $S_n = X_1 + \cdots + X_{100}$ be the total. Each X_i is uniform on $\{1, \ldots, 6\}$ with $\mu = 7/2$ and $\sigma^2 = 35/12$. By the CLT,

$$\frac{S_{100} - 350}{\sqrt{100 \cdot 35/12}} \approx \mathcal{N}(0, 1).$$

So $\mathbb{P}(S_{100} \leq 330) \approx \Phi\left(\frac{330-350}{\sqrt{2916.67/100}}\right) = \Phi(-1.17) \approx 0.121$.

LLN and Monte Carlo

The law of large numbers justifies Monte Carlo estimation: the sample average of rewards converges to the expected return. This is why Monte Carlo policy evaluation works. The CLT tells us how fast: the error is approximately normal with standard deviation proportional to $1/\sqrt{n}$, giving us confidence intervals for value estimates.

12 Tail Bounds and Confidence Intervals

Chebyshev's inequality gives tail bounds in terms of the variance, but these are often loose. For bounded random variables, we can do much better.

12.1 Chebyshev Confidence Intervals

Proposition 12.1 (Chebyshev Confidence Interval)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. For any $\delta \in (0, 1)$,*

$$\mathbb{P}\left(|\bar{X}_n - \mu| < \frac{\sigma}{\sqrt{n\delta}}\right) \geq 1 - \delta.$$

Proof. By Chebyshev's inequality with $k = \sigma/\sqrt{n\delta}$:

$$\mathbb{P}\left(|\bar{X}_n - \mu| \geq \frac{\sigma}{\sqrt{n\delta}}\right) \leq \frac{\text{Var}(\bar{X}_n)}{k^2} = \frac{\sigma^2/n}{\sigma^2/(n\delta)} = \delta.$$

Taking complements gives the result. □

12.2 Hoeffding's Inequality

Lemma 12.2 (Hoeffding's Lemma)**.** *If X is a random variable with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely, then for all $t > 0$,*

$$\mathbb{E}[e^{tX}] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

Proof. Since $a \le X \le b$, we can write $X = \alpha b + (1-\alpha)a$ where $\alpha = \frac{X-a}{b-a} \in [0,1]$. By convexity of e^{tx}:

$$e^{tX} \le \frac{X-a}{b-a}e^{tb} + \frac{b-X}{b-a}e^{ta}.$$

Taking expectations and using $\mathbb{E}[X] = 0$:

$$\mathbb{E}[e^{tX}] \le \frac{-a}{b-a}e^{tb} + \frac{b}{b-a}e^{ta}.$$

Let $p = \frac{-a}{b-a}$, $h = t(b-a)$. Then

$$\mathbb{E}[e^{tX}] \le p\, e^{(1-p)h} + (1-p)\, e^{-ph} = e^{-ph}\left(pe^h + 1 - p\right).$$

Define $\varphi(h) = -ph + \ln(pe^h + 1 - p)$. Then $\varphi(0) = 0$ and $\varphi'(0) = 0$ (verify: $\varphi'(h) = -p + \frac{pe^h}{pe^h+1-p}$, and at $h = 0$ this gives $-p + p = 0$). Furthermore,

$$\varphi''(h) = \frac{pe^h(1-p)}{(pe^h + 1 - p)^2} \le \frac{1}{4},$$

since $\frac{u(1-u)}{1} \le 1/4$ for $u \in [0,1]$, applied to $u = \frac{pe^h}{pe^h+1-p}$. By Taylor's theorem, $\varphi(h) = \varphi(0) + \varphi'(0)h + \frac{1}{2}\varphi''(c)h^2 \le h^2/8$ for some c. Therefore $\mathbb{E}[e^{tX}] \le e^{h^2/8} = e^{t^2(b-a)^2/8}$. □

Theorem 12.3 (Hoeffding's Inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables with $a_i \le X_i \le b_i$ almost surely. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then for any $\varepsilon > 0$,*

$$\mathbb{P}\big(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \ge \varepsilon\big) \le 2\exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Proof. Let $S_n = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[S_n]$. We bound $\mathbb{P}(S_n - \mu \ge n\varepsilon)$; the lower tail is symmetric.

For any $t > 0$, by Markov's inequality applied to $e^{t(S_n-\mu)}$:

$$\mathbb{P}(S_n - \mu \ge n\varepsilon) = \mathbb{P}(e^{t(S_n-\mu)} \ge e^{tn\varepsilon}) \le \frac{\mathbb{E}[e^{t(S_n-\mu)}]}{e^{tn\varepsilon}}.$$

By independence, $\mathbb{E}[e^{t(S_n-\mu)}] = \prod_{i=1}^n \mathbb{E}[e^{t(X_i - \mathbb{E}[X_i])}]$. Each centered variable $X_i - \mathbb{E}[X_i]$ satisfies $a_i - \mathbb{E}[X_i] \le X_i - \mathbb{E}[X_i] \le b_i - \mathbb{E}[X_i]$, a range of $b_i - a_i$. By Hoeffding's Lemma, $\mathbb{E}[e^{t(X_i - \mathbb{E}[X_i])}] \le e^{t^2(b_i-a_i)^2/8}$. Therefore

$$\mathbb{P}(S_n - \mu \ge n\varepsilon) \le \exp\left(-tn\varepsilon + \frac{t^2}{8}\sum_{i=1}^n (b_i - a_i)^2\right).$$

Minimizing over t: the exponent is minimized at $t = \frac{4n\varepsilon}{\sum_{i=1}^n (b_i-a_i)^2}$, giving

$$\mathbb{P}(S_n - \mu \ge n\varepsilon) \le \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

The same bound holds for $\mathbb{P}(S_n - \mu \le -n\varepsilon)$ by applying the argument to $-X_i$. A union bound gives the two-sided result. □

Corollary 12.4 (Hoeffding for i.i.d. Bounded Variables)**.** *If $X_1, \ldots, X_n$ are i.i.d. with $a \le X_i \le b$, then*

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le 2e^{-2n\varepsilon^2/(b-a)^2}.$$

Proof. Set $a_i = a$ and $b_i = b$ for all i in Theorem 12.3. Then $\sum_{i=1}^n (b_i - a_i)^2 = n(b-a)^2$. □

12.3 Sample Complexity

Proposition 12.5 (Sample Complexity from Hoeffding)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with $a \le X_i \le b$. To guarantee $\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le \delta$, it suffices to take*

$$n \ge \frac{(b-a)^2}{2\varepsilon^2} \ln \frac{2}{\delta}.$$

Proof. By Corollary 12.4, we need $2e^{-2n\varepsilon^2/(b-a)^2} \le \delta$. This is equivalent to $e^{-2n\varepsilon^2/(b-a)^2} \le \delta/2$, i.e., $-2n\varepsilon^2/(b-a)^2 \le \ln(\delta/2) = -\ln(2/\delta)$. Rearranging: $n \ge \frac{(b-a)^2}{2\varepsilon^2} \ln \frac{2}{\delta}$. □

Remark 12.1. The dependence $n = O(\varepsilon^{-2} \ln \delta^{-1})$ is the standard sample complexity for Monte Carlo estimation of a bounded mean. The logarithmic dependence on $1/\delta$ is much better than Chebyshev's $O(\varepsilon^{-2}\delta^{-1})$ bound.

Sample Complexity in RL

In RL, if rewards are bounded in $[0, 1]$, the sample complexity of Monte Carlo policy evaluation with accuracy ε and confidence $1-\delta$ is $n \ge \frac{1}{2\varepsilon^2} \ln \frac{2}{\delta}$. For $\varepsilon = 0.01$ and $\delta = 0.05$, this gives $n \ge 18{,}445$ samples. Hoeffding's inequality makes this precise; Chebyshev would require $n \ge 1/(\varepsilon^2\delta) = 200{,}000$ — an order of magnitude worse.